



Optical Modeling and Optimization of Perovskite Solar Cells

*A Transfer Matrix Method (TMM) Computational
Approach*

Authors:

Viswanadhapalli Sujay (B22CS063)

M Suresh (B22CS028)

Indian Institute of Technology, Jodhpur

Design Credit Course Project

Instructor: Dr. Ravi Prakash Srivastava

Teaching Assistant: Nishant Agarwal

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Furthermore, working on this project has deeply broadened our perspective on the critical importance of solar cells and renewable energy technologies. It made us acutely realize the urgent need for India's "going green" initiatives and the vital role of long-term energy security. As our natural resources are rapidly depleting due to a heavy reliance on the burning of fossil fuels and petroleum, transitioning to advanced photovoltaic technologies like perovskite solar cells is no longer just an academic pursuit, but an absolute necessity for a sustainable and secure future.

Viswanadhapalli Sujay

M Suresh

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Abstract

The transition toward renewable energy has placed perovskite solar cells at the forefront of photovoltaic research due to their remarkable light-harvesting capabilities. While traditional material science relies heavily on empirical fabrication, our group, leveraging a computer science background, engineered a programmatic pipeline to simulate the optical behavior of these devices. We implemented the Transfer Matrix Method (TMM) in Python to analyze a predefined perovskite stack (FTO/ETL/Perovskite/HTL/Metal). By processing raw absorption coefficients into complex refractive indices, our computational model successfully mapped the internal optical interference. Our calculations verified that the 1200 nm active layer achieves over 90% absorptance across the visible spectrum, yielding a theoretical maximum photocurrent (J_{max}) of 41.22 mA/cm². We also mapped the internal electric field intensity ($|E|^2$) to confirm optimal energy localization. This work successfully bridges optical parameter extraction with downstream machine learning and electrical simulations.

1 Introduction

Global energy demands heavily necessitate the transition to sustainable sources. Among third-generation photovoltaics, organic-inorganic halide perovskites have shown unprecedented growth in Power Conversion Efficiency (PCE). However, optimizing a solar cell is a multi-dimensional problem. Before electrical simulation tools like SCAPS-1D can model charge transport, it is mathematically required to understand how light actually propagates, interferes, and decays within the ultra-thin semiconductor layers.

Our project objective was to handle the optical modeling phase. Because the layers in a perovskite solar cell are on the nanometer scale—comparable to the wavelength of incident light—simple Beer-Lambert absorption principles fail to capture the reality of standing waves and internal reflections. To solve this, we implemented the Transfer Matrix Method (TMM) algorithm programmatically. This report details our methodology in processing raw device data, simulating the optical physics, and extracting vital efficiency limits to inform the broader design credit project.

2 Device Architecture

The reference device structure provided for our simulation comprises five distinct layers. The incident light passes through the transparent window and penetrates the functional layers before hitting the back metal contact.

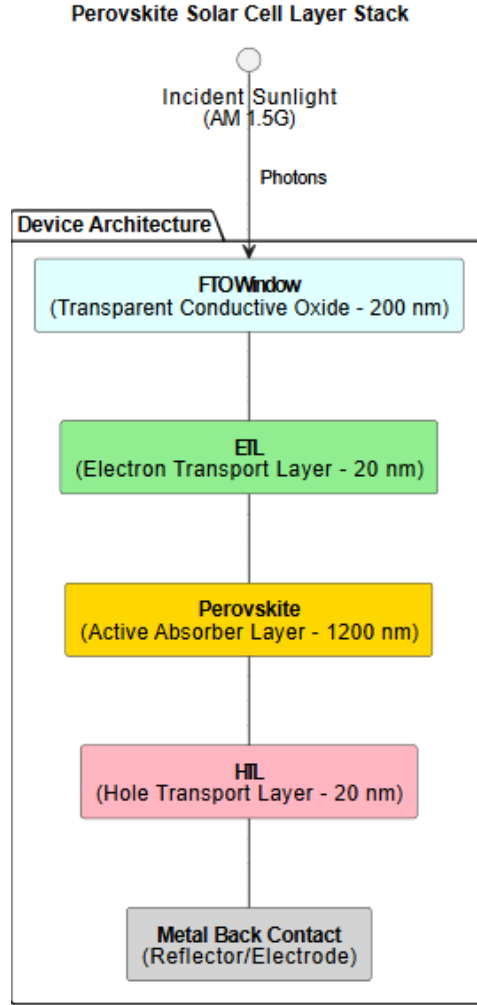


Figure 1: Schematic representation of the modeled Perovskite Solar Cell stack.

- **FTO (200 nm):** Fluorine-doped Tin Oxide acts as the transparent conductive front electrode.
- **ETL (20 nm):** The Electron Transport Layer restricts the flow of positive holes while facilitating electron transfer.
- **Perovskite (1200 nm):** The active absorber layer responsible for generating electron-hole pairs upon photon absorption.
- **HTL (20 nm):** The Hole Transport Layer restricts electrons and conducts holes to the metal contact.
- **Metal:** Acts as both the rear electrical contact and an optical mirror, reflecting unabsorbed light back into the device.

3 Methodology: Computational Optical Modeling

3.1 Data Engineering and Optical Constants

The initial dataset provided via SCAPS-1D output contained only the absorption coefficients (α) for the layers. TMM algorithms inherently require the complex refractive index ($\tilde{N}$) to compute wave interference. Coming from a computer science perspective, our first task was to construct a clean, computationally ready dataset.

We mathematically derived the wavelength-dependent extinction coefficient (k) using the relation:

$$k(\lambda) = \frac{\alpha(\lambda) \cdot \lambda}{4\pi} \quad (1)$$

Since SCAPS evaluates electrical properties, the real refractive indices (n) were omitted from the raw data. We compiled the missing n values from established optical literature matching our exact materials to complete the required parameters over the 300 nm to 900 nm spectral grid.

3.2 The Transfer Matrix Method (TMM)

We utilized Python to automate the TMM linear algebra sequence. TMM treats the electric field of light as a 2D vector comprising forward (E^+) and backward (E^-) traveling waves. For each layer boundary, the code builds a dynamical interface matrix based on the Fresnel reflection coefficient (r):

$$r = \frac{\tilde{N}_1 - \tilde{N}_2}{\tilde{N}_1 + \tilde{N}_2} \quad (2)$$

Simultaneously, for propagation through a layer's thickness (d), the wave undergoes a phase shift (δ) defined by:

$$\delta = \frac{2\pi\tilde{N}d}{\lambda} \quad (3)$$

By computing the dot product of all interface and propagation matrices across the multi-layer stack, our script determines the total fraction of light reflected (R), transmitted (T), and absorbed (A) per layer.

4 Results and Analysis

The fully reproducible Google Colab environment containing our code and dynamic simulations is accessible here: [Project TMM Notebook](#).

4.1 Optical Absorptance and Parasitic Losses

The primary output of the TMM simulation is the layer-by-layer optical absorption profile.

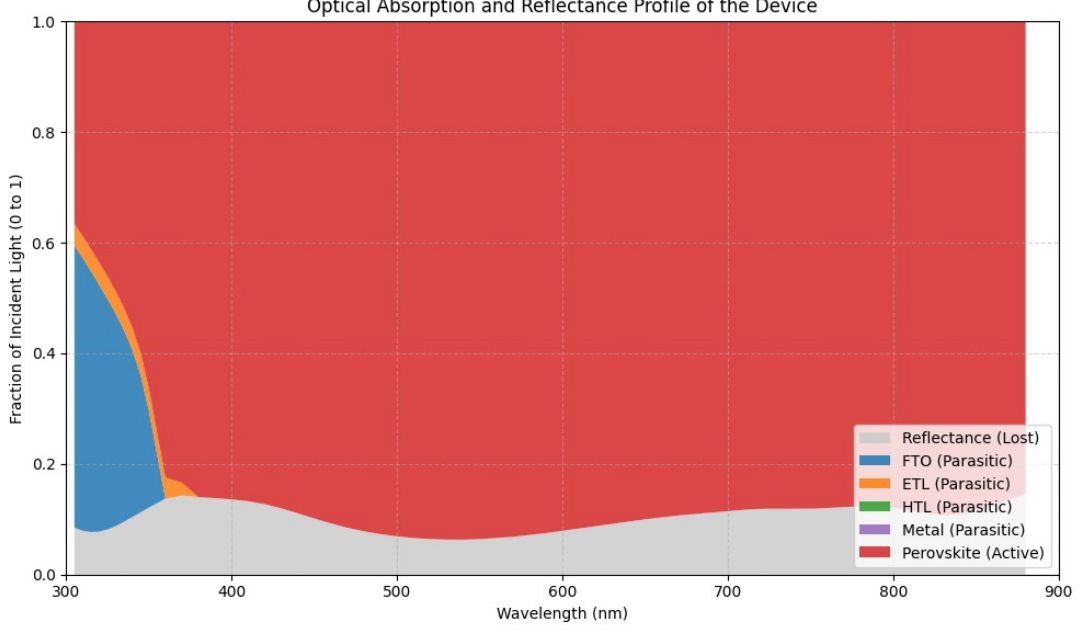


Figure 2: Simulated fraction of incident light absorbed by each layer.

As seen in Figure 2, the 1200 nm active perovskite layer demonstrates exceptional light harvesting, absorbing over 90% of incident light across the visible spectrum (400 nm – 750 nm). However, our model identified a significant parasitic loss in the Ultraviolet regime (< 400 nm), where the FTO glass acts as a bottleneck by absorbing light before it can reach the active layer, thereby generating unwanted heat rather than excitons.

4.2 Theoretical Photocurrent Generation

Optical absorptance does not directly equal electrical efficiency. To quantify the physical potential of the cell, we integrated our TMM absorptance data (A) against the standard AM 1.5G solar photon flux (Φ). Assuming 100% internal quantum efficiency (IQE), we calculated the maximum theoretical short-circuit current density (J_{max}):

$$J_{max} = \int_{\lambda=300}^{\lambda=900} q \cdot \Phi(\lambda) \cdot A_{perovskite}(\lambda) d\lambda \quad (4)$$

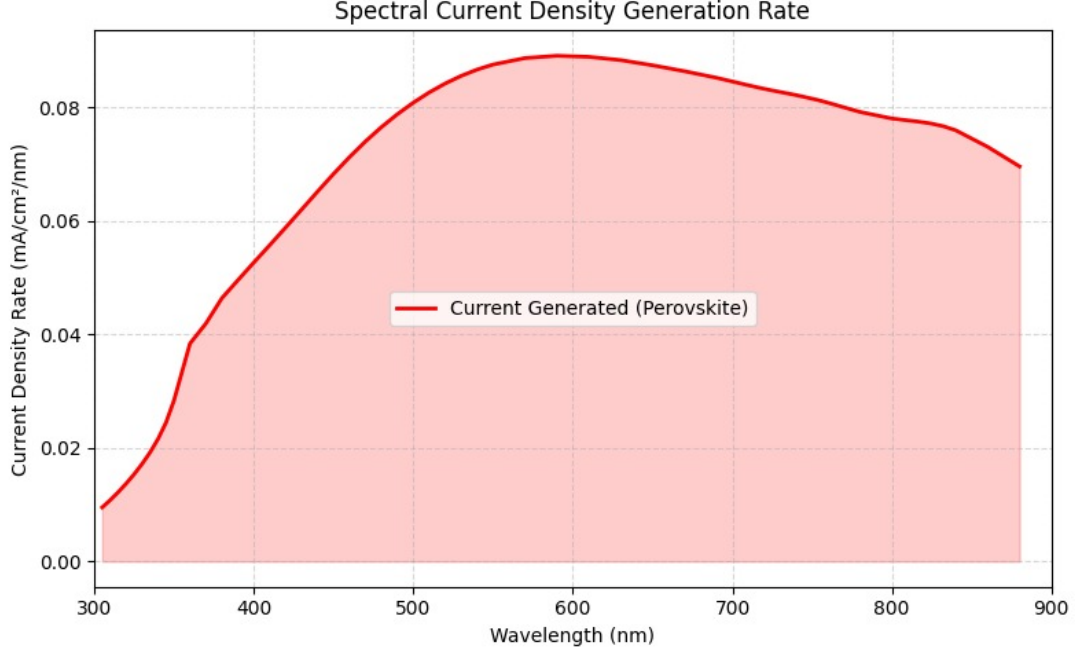


Figure 3: Spectral current density generation rate under AM 1.5G illumination.

The computational integration yielded a J_{max} of **41.22 mA/cm²**. This represents the absolute physical speed limit of the cell's current generation capability based entirely on the geometry and optical trapping of the chosen thicknesses.

4.3 Internal Optical Electric Field Profile

Finally, to verify the spatial dynamics of the interference standing waves, we computed the normalized optical electric field intensity ($|E(x)|^2$) as a function of device depth.

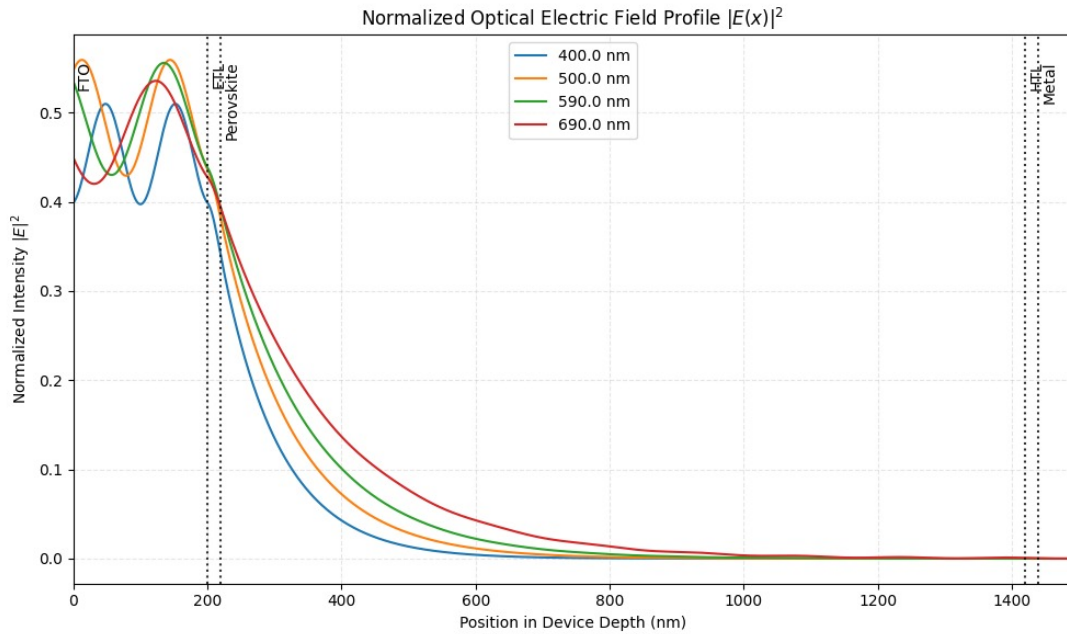


Figure 4: Normalized optical electric field profile across the device depth.

Figure 4 illustrates the constructive and destructive interference waves. Critically, the algorithm proves that the highest energy intensity peaks are safely localized within the boundaries of the 1200 nm perovskite layer. The field rapidly decays within the HTL and ETL transport layers, confirming that the structural dimensions are highly optimized to trap light exactly where charge separation occurs.

5 Conclusion

This design credit project successfully bridged the domains of computational linear algebra and optical materials science. By building an automated Transfer Matrix Method pipeline, we accurately mapped the nanometer-scale optical interference within a multilayer perovskite solar cell. We transformed raw absorption coefficients into a robust dataset, calculated the precise locations of parasitic losses, and extracted a highly competitive theoretical photocurrent limit of 41.22 mA/cm².

From a computer science perspective, the automation of these complex wave equations provided a vital missing link. The highly accurate optical datasets and theoretical limits established by our pipeline are now perfectly formatted to serve as foundational inputs for the next phase of the project: electrical simulation and Machine Learning-driven structural optimization.

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